Week 7 Progress Report

ECE501: Digital Image Processing | GROUP 19 | November 1, 2026
Project Topic: Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors
Deliverable: Corrected evaluation setup, edge-band handling, Gabor features, threshold selection, window-size study, LBP+Gabor fusion, score-based image-level rule, and rotation test

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Objectives of the Week

- Fix the evaluation setup (data handling, calibration set, threshold) identified as the main problem in Week 6.
- Handle the edge band.
- Add Gabor features and compare them fairly with LBP.
- Test window size, LBP+Gabor fusion, a score-based image-level rule, and rotation.

Not done this week and moved to Week 8: GLCM with ground-truth masks, AND/OR fusion on the new split, cTPR, the published-AITEX comparison, cross-validation over strips, per-defect-type recall, the two-class model comparison, and 8x32 / 32x8 windows. **[Add one line of reason: this week went into diagnosing the setup and the edge band.]**

Work Done This Week

A. Evaluation setup (changed from Week 6)

- Fabric type 03 images are 256 x 4096 strips. Week 6 resized them to 256 x 256, squeezing them about 16x horizontally. This week each strip is cut into 256 x 256 tiles with no resizing, so defects keep their real size.
- 20 defect-free strips (320 tiles) and 30 defective strips with masks (31 exist, 1 has no mask).
- Defect-free strips were split by source strip, not by tile, to avoid leakage: 12 strips for the reference, 4 for threshold calibration, 4 for normal testing. Defective strips were split by strip into validation and test halves.
- Tiles are the units for "image-level" results, so Week 6 and Week 7 image-level numbers are not directly comparable.
- Mean defect size per defective tile: 2510 pixels. **[Add median, min and max from the one-line check.]**

B. Edge band: diagnosis and fix

- With the new split, LBP image AUC was 0.413 (below chance) and nothing was flagged (threshold 0.565).
- Mean intensity by tile index showed tiles 0 to 3 are pure white (mean exactly 255) in every strip, and tile 4 is partly white (mean about 110 to 122 vs about 87 for normal fabric).
- Excluding tiles 0 to 3 lost 0 of 220,886 defect pixels. LBP AUC rose to 0.672.
- Tile 4 holds 11,348 defect pixels (5.1%). Dropping it as well was tested later (Section F).

C. LBP vs Gabor (tiles 4 to 15, 99th-percentile threshold, 12-pixel rule)

Metric	**LBP**	**Gabor**
Image AUC (max)	0.672	0.852
Pixel AUC	0.703	0.884
Image TP/FN/FP/TN	8/80/2/46	13/75/4/44
Pixel IoU / Dice	0.0000 / 0.0000	0.0011 / 0.0023

Gabor setup: 12 filters (3 wavelengths x 4 orientations), mean and std of the magnitude per 16 x 16 patch (24 features), Mahalanobis distance from the normal reference.

D. Threshold selection (tiles 4 to 15, test half only)

The threshold percentile was chosen on a validation half of the defective strips (by Dice), then reported on the held-out test half.

Method	**Week 6 rule (99th pct)**	**Chosen on validation**
Gabor	Dice 0.003	97th pct: precision 0.101, recall 0.690, IoU 0.096, **Dice 0.176**
LBP	Dice 0.000	95th pct: precision 0.039, recall 0.350, IoU 0.036, **Dice 0.069**

Gabor pixel AUC on the test half is 0.925 (LBP 0.723). The validation set is small (14 strips), so the chosen percentile may be unstable.

E. Window size and fusion (Gabor, tiles 4 to 15, test half)

Patch size	**8**	**16**	**32**
Dice	0.132	0.176	0.192
Pixel AUC	0.927	0.925	0.912

Patch size did not fix localisation, and 16 vs 32 is within noise. Smaller square windows did not help, so the Week 6 suggestion was not supported for Gabor. Thin windows were not tested. Score-level LBP+Gabor fusion (LBP weight 0, 0.25, 0.5) gave Dice 0.1755, 0.1749 and 0.1743,

so there was no gain at these hand-picked weights.

F. Excluding the mixed tile 4 (tiles 5 to 15, Gabor, test half)

Metric	With tile 4	Without tile 4
Chosen percentile	97	99.5
Precision / Recall	0.101 / 0.690	0.276 / 0.749
IoU / Dice	0.096 / 0.176	**0.252 / 0.403**
Pixel AUC	0.925	0.960
Normal pixels flagged	2.7%	0.6%

- The gain comes mostly from the calibration threshold: the 99th-percentile threshold fell from 15.14 to 7.46 once tile 4 left the calibration set. At the 97th percentile, Dice is unchanged (0.176 vs 0.177).
- The test set also changed (it lost its hardest tiles), so part of the gain may come from an easier test set. **[A paired check on the old model is pending.]**
- 99.5 is the top of the tested grid, so it may not be optimal.
- Cost: 5.1% of defect pixels are not inspected.

G. Image-level rule

Instead of "at least 12 flagged pixels", a tile is called defective if its maximum score exceeds a validation-chosen threshold.

- Tiles 4 to 15: recall 0.576 (19/33), false-alarm rate 0.083 (4/48), AUC 0.784. The old rule caught 14.8% at the same false-alarm rate.
- Tiles 5 to 15: TP=16, FN=11, FP=0, TN=44, recall 0.593, AUC 0.880. Zero false alarms in 44 tiles does not mean zero in general.

H. Rotation test (tiles 5 to 15, thresholds fixed, no refitting)

Metric	0 deg	90 deg	180 deg
Image AUC	0.880	0.352	0.880
Dice	0.403	0.015	0.403
Normal pixels flagged	0.6%	100%	0.6%

The detector is invariant to 180 degrees but fails at 90 degrees, where every pixel is flagged. Our interpretation (untested) is that the orientation channels swap against a reference fitted on upright fabric. 270 degrees was not run and no rotated reference was tried. On a real inspection line the fabric orientation is fixed, so this matters mainly after any change of camera or fabric orientation.

What this week shows

1. The Week 6 failures were mostly setup problems: resizing, a tiny calibration set, and the

white edge tiles. They were not only problems of the descriptors.
2. Gabor ranks defects much better than LBP (pixel AUC 0.925 vs 0.723).
3. The threshold rule was the main bottleneck, and the mixed tile 4 was inflating the calibration scores.
4. Window size and fusion at the tested settings did not help.
5. Localisation is still modest: Dice 0.40 at best on tiles 5 to 15, with 5.1% of defect pixels left uninspected.

Changes from Week 6

Week 6	**Week 7**
Resized strips, 10 test images	256 x 256 tiles, group-based split, 44 normal and 27 defective test tiles
99th-percentile threshold, 12-pixel rule	Validation-chosen percentile, score-based image rule, AUC reported
Edge band suspected	Diagnosed (white tiles), excluded, cost measured
Gabor planned	Done; beats LBP
Rotation, window sweep planned	Done (square windows only)

Plan for Next Week (Week 8)

- GLCM with masks (IoU and Dice), stating exactly what Enhanced_KNN and Enhanced_Mahalanobis add (still a placeholder in Week 6).
- A paired comparison of the old and new test sets for the tile-4 result.
- AND/OR fusion and a tuned fusion weight on the new split.
- Thin windows (8x32, 32x8), cross-validation over strips, and per-defect-type recall.
- cTPR, the published-AITEX comparison, and the two-class model on the same split.
- A larger validation set, so the threshold choice is more stable.

Conclusion

[Write 3 to 4 sentences from "What this week shows": setup fixes, Gabor over LBP, Dice 0.403 with its caveats, rotation limit.]

References

[Paste references 1 to 7 from the Week 6 report here, and add one for Gabor texture features if you cite it.]